

Advanced Machine Learning

DATA 642

Lecture 4 — Sparsity Aware Learning

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, and Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier.

Bias-variance trade-off

Simple linear regression model

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_l x_l + \epsilon \quad (1)$$

We can improve estimation estimation of st. errors by replacing plain least squares fitting with some alternative fitting procedure.

Goal

Get better *accuracy* and improve *interpretability*

Prediction accuracy

1. If $N \gg l$ and provided that true relationship between y and x s is approximately linear then least squares estimates tend to have low variance and low bias.
2. If $N > l$ (not much larger than l) and relationship between y and x is approximately linear we could have a lot of variability in the least squares fit, resulting in overfitting and poor predictions. A small change in the training data can cause a large change in the least squares coefficient estimates.
3. If $l > N$ then there is no longer a unique least squares coefficient estimate. Variance is "infinite".

Model interpretability

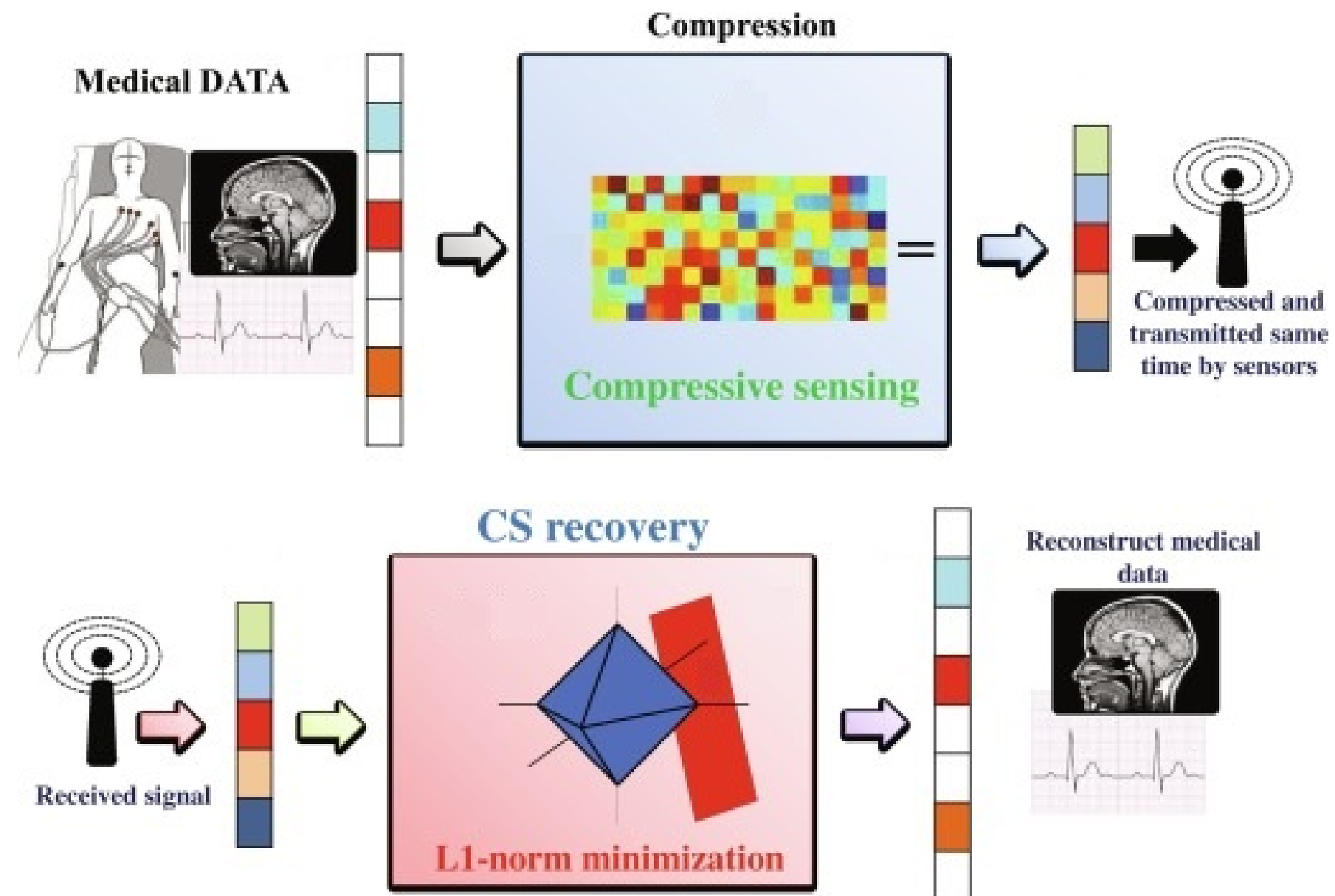
As flexibility increases interpretability could decrease. Including "irrelevant" variables into a multiple regression model leads to unnecessary complexity in the resulting model. We could remove variables by setting them equal to zero.

Solution

Constrain or shrink the estimating coefficients so we can reduce variance while increase a little bit bias in order to improve accuracy.

Compressed Sensing (CS)

- The goal is to directly acquire few samples as possible that encode the minimum information needed to obtain a compressed signal representation.



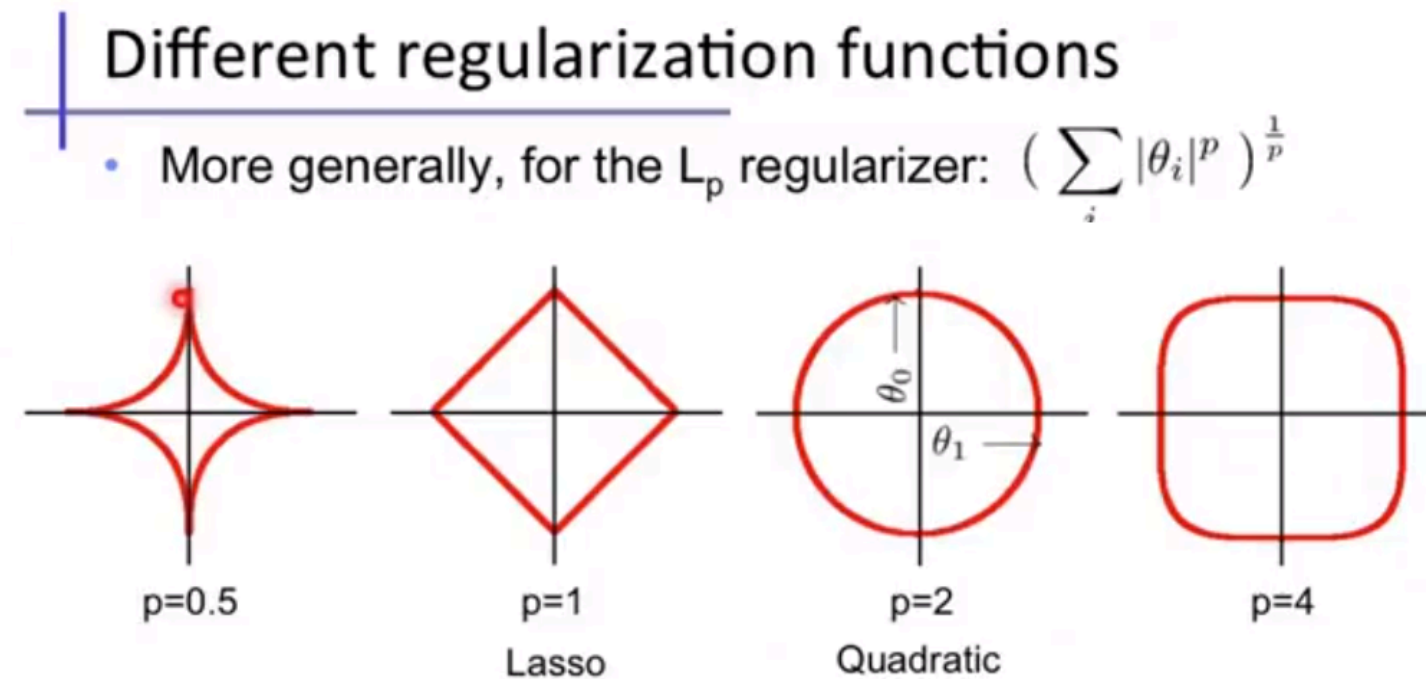
- Let $\mathbf{X} \in \mathbb{R}^{N \times l}$ sensing matrix, which is applied to the unknown signal vector $\mathbf{s} \in \mathbb{R}^{l \times 1}$ in order to obtain observations $\mathbf{y} \in \mathbb{R}^{N \times 1}$ and let Ψ be a dictionary matrix that describes the domain where $\mathbf{s}$ accepts a **sparse representation**

$$\mathbf{s} = \Psi \boldsymbol{\theta} \quad (2)$$

- From $\mathbf{y} = \mathbf{X}\mathbf{s}$ we obtain that $\mathbf{y} = \mathbf{X}\Psi\boldsymbol{\theta}$ given $\mathbf{X}$, Ψ or $\mathbf{X}\Psi$. It is possible to get $\hat{\mathbf{s}}$ by solving an optimization problem using the ℓ_1 -norm and assuming that at most k of $\boldsymbol{\theta}$ components are non-zero.
- At the heart of the CS problem lies an under-determined set of linear equations, which in general, accepts an **infinite number of solutions**.
- The extra piece of information is that the true model whose estimate we want to obtain is **sparse**, that is only few of its coordinates are non-zero.

ℓ_0 -norm to promote sparsity

- An interesting norm to incorporate and measure sparsity is the ℓ_0 -norm which adds the non-zero components of a vector.



- We deal with optimization so derivative is **not** defined and dealing with the ℓ_0 -norm is an NP-problem.
- Dealing with different ℓ_p -norms is also **not** convenient because problem becomes non-convex so we have challenges with optimization.

Practical Examples

- **Natural Language Processing (NLP):** Many words in a document may have zero occurrences or very low frequencies, leading to sparse representations. For example, in text classification tasks, documents are often represented as sparse vectors where each dimension corresponds to a unique word in the vocabulary.
- **Image Processing:** In image processing, most pixels in an image are typically background and carry little useful information. Sparse representations can be obtained by focusing only on the relevant features or regions of interest, leading to more efficient storage and computation.

Emerging Applications

- **Medical Imaging:** Sparsity-aware learning techniques are increasingly used in medical imaging for identifying relevant biomarkers and features associated with diseases. For example, sparse MRI techniques aim to acquire fewer measurements while still accurately reconstructing images, reducing scan times and patient discomfort.
- **Signal Processing:** In signal processing, sparsity-aware methods such as compressive sensing are used for tasks like denoising and signal reconstruction. These techniques exploit the sparse nature of signals to efficiently recover them from undersampled measurements, leading to improved signal quality and reduced data acquisition time.
- **Recommendation Systems:** Sparsity-aware methods are applied in recommendation systems to address the cold start problem and improve recommendation accuracy. By leveraging sparse user-item interaction data, these methods can identify relevant items for users even with limited or incomplete information, leading to more personalized recommendations.